Ayaan Yousaf

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TECHNICAL SKILLS

Languages: Java, Python, JavaScript, C++, HTML, CSS
Frameworks/Libraries: React.js, Vue.js, Node.js, Express.js, Tailwind CSS, PyTorch, NumPy, pandas, Matplotlib
Tools: Git, VS Code, IntelliJ, Jupyter Notebook, MongoDB, PostgreSQL, Vercel, RESTful APIs

EDUCATION

Bachelor of Science (Honours) Co-op in Computer Science — **Oshawa, Ontario**
Ontario Tech University — *Expected Graduation, May 2027*

- **GPA:** 3.68/4.3, *President's List for Winter 2023 and Fall 2024*
- **Related Coursework:** Data Structures, Algorithms, Computer Architecture, Database Systems, AI-Powered Software Engineering, Software Systems Development, Software Design, Web Development, Machine Learning

PROJECTS

Plagued Abyss | Godot, GDScript, OOP, Systems Design — *Mar 2026 – Apr 2026*

- Developed a 2D wave-based survival game in Godot featuring 10+ interconnected systems including progression, combat, level design, economy, and difficulty-scaling.
- Implemented AI pathfinding for enemies using a navigation mesh in Godot, enabling enemies to avoid obstacles.
- Refined collision detection, enemy spawning, and combat mechanics through iterative playtesting and debugging.
- Designed and implemented levels and core gameplay systems from initial concepts to a fully playable release.

DRL Software Testing Framework | Python, Stable-Baselines3, Gymnasium, Selenium — *Sept 2025 – Oct 2025*

- Engineered a modular Deep Reinforcement Learning framework to automate web app and game testing using agents trained on PPO and A2C algorithms.
- Designed and evaluated 4 custom reward functions for simulating distinct user behaviours during testing.
- Accelerated agent training speeds by 6x through environment parallelization with SubprocVecEnv.
- Achieved 95% reusability by decoupling environment logic, training/evaluation scripts, and YAML configuration.

Portfolio Website | React.js, Three.js, Vite, Tailwind CSS, Vercel — *Aug 2025 – Sept 2025*

- Built and deployed a responsive portfolio on Vercel, featuring 10+ reusable React components and custom hooks.
- Implemented smooth scroll-triggered animations with Framer Motion, enhancing user experience and interactivity.
- Integrated a dynamic 3D background and interactive Earth model using Three.js.
- Improved accessibility by ensuring compatibility across multiple devices for a consistent user experience.

BookNest | Vue.js, D3.js, Node.js, Express, MongoDB, RESTful APIs — *Mar 2025 – Apr 2025*

- Developed a full-stack digital library web application using Node.js, Vue.js, Express, and MongoDB.
- Modeled schemas for core entities using Mongoose and automated database seeding with Node.js scripts.
- Connected frontend Vue.js web pages with backend REST APIs to dynamically retrieve book data.
- Debugged UI issues and implemented advanced search filters, ensuring smoother navigation and usability.

WORK EXPERIENCE

RK Studios — **Toronto, Ontario**
IT Support Technician — *Aug 2021 – Aug 2022*

- Configured hardware and software efficiently for wedding photography, allowing for operation on schedule.
- Managed IT systems and software, including Adobe Photoshop and Lightroom.
- Provided technical troubleshooting, resolving issues with Google Drive and communicating clear solutions to staff.

ACTIVITIES AND CERTIFICATES

Computer Science Club – Ontario Tech University — **Oshawa, Ontario**
General Member — *Sept 2023 – Present*

- Participated in a Microsoft-sponsored hackathon event focused on Azure cloud services.